



Research article

Exploring the Chatbot usage intention-A mediating role of Chatbot initial trust

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ABSTRACT

Chatbots, an integral part of Digitalization, are becoming the cornerstone of modern-day marketing practices. This study's main objective is to identify the Chatbot traits that have a bearing on users' confidence, acceptability, and engagement. Chatbots hold much potential for customer support in terms of service efficiency, cost savings, and customer experience. In addition to saving business costs, chatbots improve customer engagement, by reducing time and effort they need to spend in doing their business transactions. Aspiring, well-educated, and tech-savvy Gen Z customers perceive chatbots as highly compatible with their preferred method of seeking product-related information. As chatbots play a significant role in the marketing process, this study aims to identify the chatbot traits that influence customers' confidence, acceptance, and engagement with chatbots. Using the PLS-SEM tool/technique we computed a structural equation measurement model to test the traits leading to Chatbot trust. The study also found that perceived ease of use, performance expectations, and social influence of chatbots are all greatly enhanced by trust on the part of the customers. Detailed statistical analysis showed that the trust that consumers have in chatbots strongly influences consumers' intentions, loyalty, and engagement levels. While gender had no impact on consumer intentions to use chatbots, the age of the respondents played a substantial role in their plans to use chatbots.

1. Introduction

Many companies have adopted chatbots to enhance customer services. Companies see them as an innovative touchpoint that offers conversations with prospective customers as well as existing customers [1]. According to Ref. [2] Chatbots are computational systems capable of conversing with humans through natural language. Chatbots have recently become very popular due to technological advancements and the fact that texting has replaced face-to-face interactions as the principal method of personal and professional communication [3]. When traditional marketing is insufficient, artificial intelligence is thought to be able to improve customer interactions [4]. A chatbot may assist in being proactive and personable in company communications with customers by promptly and proactively answering customer inquiries [5]. Chatbots can be meshed with numerous communication channels like social media platforms, company websites, and even messaging apps as businesses perceive many potential advantages in using these technologies. For example, chatbots can respond instantaneously to queries, give consistent information, and be a source for collecting customer data. AI chatbots are helpful for various purposes and industries, including education, healthcare, customer service, and entertainment

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as they are available 24/7. Advanced chatbots can be programmed and tailored to provide more enjoyable and engaging conversations. However, like any other technology, a chatbot's effectiveness depends on continuously upgrading technology to meet evolving customer needs.

Chatbots have recently attracted much attention from researchers interested in their possible impact on customer satisfaction [1], and how customers view companies [3]. Consumers and AI-driven systems now interact with enterprises online more often than humans with one other [6]. As this study is done on Gen Z, a brief generation description is given below. Gen Z is ill-defined, according to Refs. [7,8]. Generation Z will be defined in this paper as those born between 1995 and 2012. [7], One generation that has grown up with continual access to the internet—known as Generation Z, or I Gen—has had “smart” mobile devices throughout their entire life [7]. Compared to previous generations, millennials started using cell phones and the internet as teenagers, which marks a significant change from the experiences of previous generations. Most millennials were adults when these technologies first gained traction. Gen Z is different from prior generations in many respects when it comes to their distinct upbringing, experiences with family dynamics, and the ubiquitous presence of innovative technology in their daily lives. Determining which characteristics of AI chatbots are most likely to encourage brand loyalty could help firms better engage with Generation Z. This serves as another evidence of the study's usefulness for firms. Finally, research on how AI chatbots affect loyalty across generations is rare. This implies that it may be more advantageous from a scientific perspective to concentrate on a particular generation, such as Gen Z.

According to several studies [9–11] Online trust is a significant factor in a website's capacity to compete in the marketplace and maintain enduring customer relationships. Trust is fundamental to relationships of all kinds [12]. This is especially true with e-commerce, as customers need help to assess a vendor's credibility in the same way they could in a traditional face-to-face exchange due to the restricted nature of the Web interface [10]. According to Ref. [13], customers' first trust in chatbots is considerably increased through perceived ease of use, compatibility, and social impact.

This study aims to bridge the knowledge gap by investigating the critical factors that influence initial trust in chatbots. Although much research has investigated trust in different technologies, research addressing initial trust in chatbots is scant. Here we test how consumer engagement and intention to use a chatbot are affected by a chatbot's initial level of trust. Addressing this knowledge gap, we present a primary data study collected through a questionnaire to form a model of customers' initial trust in Chabot. For this study, the authors took factors important to trust as recognized from the literature review. Keeping in mind the peculiarity of Chabot as a reciprocal device, the design of this study is as follows. First, we determine from the literature the factors that influence initial trust, especially in customer service marketing. Then we determine other factors of importance influencing the usage of chatbots.

The structure of the present study is organized as follows: First, the literature review and hypotheses development are introduced. Secondly, the Methodology and Conceptual Model are presented. After this, the results and analysis are presented followed by the discussion part of the findings. Finally, the limitations and future research suggestions are outlined.

2. Literature review

2.1. Chatbots initial trust

Customers' initial level of trust in a chatbot is critical in determining whether they will utilize it. Because chatbots are a relatively new technology, customers' perceptions of and plans for using them are greatly influenced by their trust in them [16]. [23] Scholars such as [9,30,31], and [32] have explored the significance of trust and its function in online transactions and purchasing behavior to attain desired results [16]. Scholars such as [9,30,31], and [29,32] have explored the significance of trust and its function in online transactions and purchasing behavior.

2.2. Chatbot initial trust based on antecedents from various theories

The technology acceptance model (TAM) was developed by Ref. [14], and it proposes that the perceived usefulness and perceived ease of use of the technology proposed influence the intention to use new technologies. Previous studies [15,16] based on the TAM Model consistently showed that adopting new technologies is based on perceived ease of use and usefulness. They are the essential drivers of initial trust and behavioral intention toward adopting new technologies.

Trust is included in the TAM in several distinct contexts. When it comes to examining initial trust in the context of e-commerce, the theories that are most commonly used as a theoretical basis are UTAUT [14,21], TAM [9], and DOI [32], Four components that make up this theory, according to Ref. [21] are- (i) expectation of effort, (ii) expectation of performance (iii) impact of social factors, and (iv) traits that promote communication.

The UTAUT theory (Unified Theory of Acceptance and Use of Technology) was developed by Ref. [42], It depends on four key constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions. UTAUT demonstrates that these four factors as mentioned above are the factors that account for 70 percent of the variance in use, offering more substantial predictive power than the rest of the models that examine technology acceptance [21–23].

The Diffusion of Innovation (DOI) model examines consumers' intention to adopt new technologies [14,17]. Of the five technologically related factors proposed by Ref. [18] – Relative Advantage; Compatibility; Complexity; Trainability; and Observability, the first three are the most important for technology adoption [14]. As Relative Advantage and Perceived Usefulness are similar [19], Compatibility was the only factor chosen from DOI theory in our study.

The model proposed in this study combines the TAM, Diffusion of Innovations Theory (DIT), and UTAUT. The authors have taken the constructs based on their consistency in results, as referred to in the literature review. TAM Model consistently showed that the

adoption of new technologies is based on both Perceived Ease of Use and Perceived Usefulness [20]. According to Refs. [24,25] Perceived Ease of Use (PEOU), Perceived Usefulness (PU), and Perceived Trust (PTRU) predict users' intentions to use chatbots. Newer versions of the UTAUT model introduced two additional variables based on the finding that the use of technology is determined by behavioral intention, Performance Expectation and Social Influence which impacted consumers' intentions to adopt chatbots. Therefore, these two factors were adopted from the UTAUT model. The Compatibility factor is the most critical factor in creating initial trust in technology and its adoption [26,27]. Therefore, Compatibility was chosen from the DOI Theory.

2.3. Perceived ease of use

Perceived ease of use is defined as 'The degree to which a person believes that using a specific system would be free of an effort', i.e. Perceived Ease of Use is the propensity of people to think that utilizing new technology is simple [14]. [33]. argue that increasing the Perceived Ease of Use can lessen any technology's difficulties. 'Perceived Ease of Use' refers to how well a system functions and how little additional work, expertise, or skill is needed from users [9,28,34], report that despite diligent implementation efforts, new technologies that do not support people while they carry out their duties are less likely to be well-accepted. Additionally, people must be capable of using any technology correctly, because even a small probability that they may make mistakes affects their trust in that technology [23], in our case, the chatbot [1]. claim that by highlighting perceived ease of use, service providers have the ability to swiftly remove hesitation in utilizing Internet-based services.

There is a positive correlation between perceived ease of use and trust. Studies use these two terms interchangeably since the TAM and UTAUT models explain perceived ease of use as effort expectancy [4,9]. The DOI theory states that customers' attitudes toward using new technology affect how they behave with it. Complexity, the antithesis of Perceived Ease of Use, is one of the technology-related aspects of the DOI theory, [13,35]. Hence, the apparent ease of use anticipates the likelihood of increased initial trust in chatbots. Considering this, we propose the following hypothesis.

H1. Perceived ease of use has a significantly positive impact on trust in using chatbots.

2.4. Perceived compatibility

Compatibility is defined by Ref. [37] as "the extent to which an innovation is viewed as being consistent with the current values, needs, and prior experiences of potential adopters." There is strong evidence from studies on both mobile banking [15] and internet banking [6,19] that consumers' expectations of how well the two systems work together significantly impact their propensity to embrace and utilize the two systems. Consistent with the domain-oriented integrity (DOO) theory [38], compatibility may affect consumers' initial trust in online merchants.

In addition, research on the adoption of mobile banking [15] and Internet banking [39] indicated that the customers' perceptions of compatibility positively affected customers' views of the product's usefulness and usability. People adopt innovations more quickly if they fit in with their lifestyles. People are more likely to embrace innovations that align with their values, so compatibility is crucial in the DOI model [6,40]. state that Compatibility is an important factor in building trust in services early on. Compatibility is key in getting people to trust and adopt new technologies quickly [26].

H2. Perceived compatibility significantly and positively impacts trust in using chatbots.

2.5. Performance expectancy

Performance Expectancy refers to an individual's belief in the effectiveness of implementing a particular program in enhancing their job performance. [42], People expect that online technology will aid them with certain activities like discovering what they need, making easy payments, and more, according to Ref. [28]. Further, other studies have demonstrated that people's trust in new technology is initially impacted when they use it and get its benefits. Previous studies have shown a high association between initial trust and performance expectancy in the digitalization sector [7]. Researchers contend that a critical element of chatbot startup is performance expectation. Performance Expectancy (PE), is defined as 'the degree to which the user expects that using the system will help him or her to attain gains in job performance.' According to Refs. [9,42,43] Chatbot services have the potential to improve user experience significantly. They are more likely to be used again by users who find them useful for online transactions, informational purposes, prompt responses, and valuable solutions.

The chatbot system's benefits could be offset by users believing it to be overly complicated and cognitively taxing. Effort expectation is the factor that defines the extent to which the chatbot system aids the user in accomplishing tasks with more success and efficiency. Users can enhance their productivity by reallocating the time and effort they save towards other work-related assignments.

H3. Performance expectancy has a significantly positive impact on trust in chatbots.

2.6. Social influence

Social influence, according to the UTAUT [42], is 'the degree to which an individual perceives how important it is for others to believe he or she should use the new system.' According to Ref. [4], these attitudes will influence consumers' trust in and adoption of new technology. According to Ref. [8], one of the most essential variables in generating confidence in a product or service is the level of

social support it receives. This component aids in the expression of various points of view regarding the dependability, compatibility, and utility of a product or service, which in turn assists prospective purchasers in deciding whether to utilize that product or service [7]. According to Ref. [4], social impact influences customers' trust in online channels.

According to the theory of reasoned action (TRA), what a person knows might affect their mood, whether positive or negative [36, 45]. The subjective norm in technology adoption refers to an individual's belief that their social circle, including friends and superiors, influences their usage and adoption behavior [46]. A user must follow subjective standards, also known as social influence, which are informal rules agreed upon by the user and social influencers. It is considered that the strength of a person's behavioral intention is strongly tied to the social impact of their peer group.

Consumer attitudes and behaviors are heavily influenced by social contacts, according to Ref. [40] who define social influence as the extent to which individuals of a social network influence the attitudes and actions of their peers. [11], underlined the importance of maintaining a positive self-concept, having accurate perceptions of reality, and responding correctly in their discussion on social influence. They also emphasized the need to make social relationships. Many people seek advice from their social networks when upgrading their technology, and some have discovered that the perceived social pressure of from others makes a difference.

According to Ref. [17], social influence has a favorable effect on people's intentions to adopt Internet services based on trust. Importantly [9], peer observation has a substantial impact on consumer decision-making, and consumers of allegedly socially unacceptable innovations may be compelled to withdraw from their social circle in the absence of sufficient social support.

H4. The social influence on the customer significantly influences trust in the chatbot.

2.7. Chatbot usage intention

Studies on consumer behavior have reported that consumers' perceptions of specific technologies are significantly connected with their intention to engage in trust-related actions. Trust is essential for chatbot adoption. People are more likely to trust those they see as trustworthy and honest [47]. The significance of trust in enhancing customer relationships with new technologies has been highlighted in various studies on technology adoption [48]. Additional research by Ref. [47] has verified that consumers' intentions to utilize cutting-edge technologies, such as online shopping and mobile banking, rely on their perceptions of this new technology being trustworthy [26]. has highlighted that a customer's willingness to adopt new technologies is greatly influenced by his level of trust. Our research demonstrates that the degree to which consumers initially trust internet businesses determines their propensity to use chatbots developed by those organizations [21]. identifies Behavioral Intention (BI) as a good indicator of a person's actual usage behavior. BI is 'the strength of one's intention to perform a specified behavior'. It would be challenging to gauge actual system usage for chatbot services because they are a relatively new technology type still in the adoption stage [45].

H5. Chatbot initial trust has a significantly positive influence on customer usage intention.

2.8. Customer loyalty

The modern customer anticipates being able to contact a business at any time, from anywhere, using any channel, and doing so with less effort. In today's digital world, when companies need to be always available, improving the customer experience is more important than ever.

According to Refs. [8,17], analyzing customer loyalty in the context of the rapidly changing and evolving digital world is fascinating. Due to heightened rivalry and emerging technologies, businesses are continuously facing challenges in luring and keeping customers [1]. Quick responses to queries or grievances from clients help them feel understood and appreciated [17]. Chatbot use fosters consumer confidence, which in turn boosts customer loyalty.

For decades, customer loyalty has been seen as crucial to business success. Consumer loyalty is the extent to which a consumer maintains a loyal attitude and behavior toward a particular business in the face of alternatives from other suppliers [49]. The expenses of acquiring new customers are much more than retaining current ones [2]. also discovered that a repeat customer guarantees excellent customer value. Because of this, improving the customer experience is more crucial than ever in the digital age. Therefore, it is important that companies need to always remain accessible [47].

H6. Chatbot initial trust has a significant positive influence on customers' loyalty to use a chatbot.

2.9. Customer engagement

Customer engagement can improve organizational performance and eventually lead to the development of a lasting business-customer connection [4]. 'The level of customers' motivation, brand-related, and context-dependent state of mind characterized by specific levels of cognitive, emotional, and behavioral activity in brand interactions' is what customer engagement is all about. Client Engagement is further defined by Ref. [15] as the 'mechanics of a customer's value addition to the firm, either through direct or/and indirect contribution.' Numerous customer online engagement behaviors, such as social interactions, and frequent website visits, as well as other actions such as word-of-mouth recommendations, reviews, knowledge sharing/storage, blogging, customer assistance, and legal action [5] can greatly impact the brand.

According to Ref. [43], customers who are confident in a company's offerings are inclined to engage with it favorably on the Internet, increasing the possibility that they will end up as brand ambassadors. leading to further customer involvement [38,57] study

has demonstrated that when consumers have faith in the social media brand community's context, they feel safer and are more engaged in the activities of the communities. Additionally, when two parties trust each other, interaction is more likely [17]. This study aims to determine whether consumers' early trust in chatbots might increase brand engagement. In a survey of students and their adoption of chatbots, it was seen that the perception of compatibility between students and chatbots positively influences adoption intention, highlighting the importance of compatibility [50].

H7. Chatbot initial trust has a significantly positive influence on customer engagement.

Finally, in this study, the six most important factors (namely, Compatibility, Perceived ease of use, Performance expectancy, Customer loyalty, Customer engagement, and social influence), which were drawn from the previously mentioned theories are tested as antecedents to chatbot initial trust, as shown in the (Fig-1) conceptual framework in the later part of the study.

3. Methodology

The data for this study was collected from students of select Higher Education Institutes (HEI) through a structured questionnaire (Selection of HEI, was Private Higher Education Institutions ranked as top 3 as per standards of HEI accreditation in the state of Andhra Pradesh, India) using a simple random technique. The survey was posted online using Google Forms. A total of 294 survey replies were gathered. After deleting 23 responses as they were incomplete, we had 271 responses left for our data analysis.

The participants were informed about the purpose of the study. They were also requested to forward the link to their peers for the online survey. Consent was obtained from each participant, and As this study does not involve any particular population like minors or other sensitive groups it bears little risk and the information will be used for beneficial purposes. Moreover, respondents were given the survey questionnaire after we got their consent. They could withdraw anytime they wanted. They were asked to fill in the information anonymously and were not given any incentives.

3.1. Measurement scale

The list of constructs, a Description of the construct, and the source of the measurement scale are listed in Table 1. The reliability of each construct, among Performance Expectancy, Social Influence, Chatbot Initial Trust, and Chatbot Usage Intention, loaded as more than 0.8; however, all the constructs are loaded more than the threshold value as per [51].

3.2. Conceptual model

4. Data analysis & interpretation

PLS-SEM is a statistical technique that does not rely on parametric assumptions. in contrast to covariance-based structural equation Modeling. This method is deemed appropriate once the objective is to evaluate the predictive capacity of a model. This research

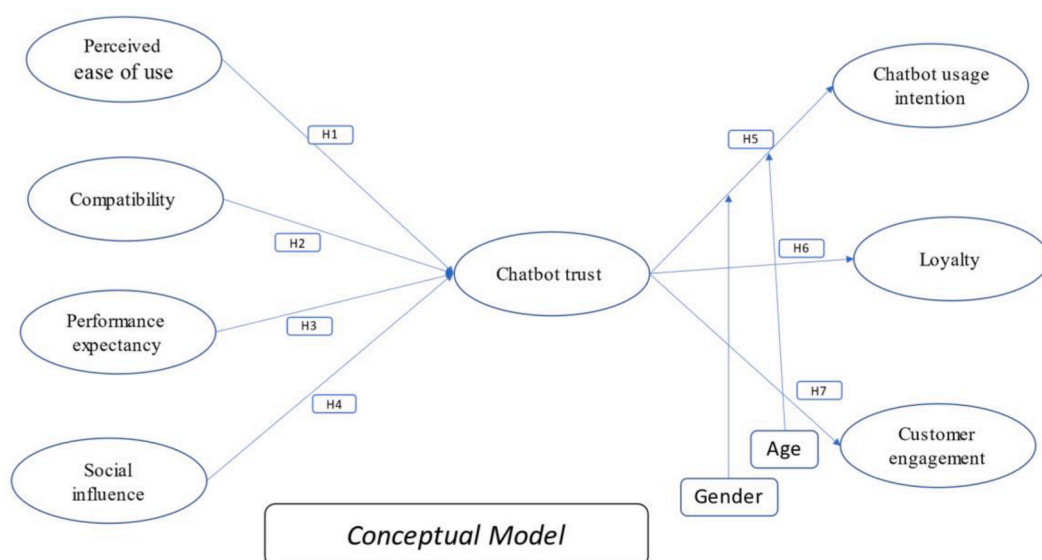


Fig.1. Conceptual Model developed by the authors.

Table 1
Sources of measurement scale.

Name of the Construct	Operational Definition	Sources of Measurement scale
Compatibility	The degree to which a novel concept or innovation conforms to the beliefs, requirements, and prior experiences of those who might embrace it is known as perceived compatibility.	[15,26]
Perceived ease of use	The propensity for people to think that a new technology would be simple to use is known as perceived ease of use.	[14].
Performance expectancy	The degree of assurance an individual exhibits that the execution of a particular activity will result in enhanced work performance.	[42]
Social influence	An individual's sense of how strongly others think they should utilize the new method.	[42]
Chatbot initial trust	Establishing trust at the outset of a relationship is not contingent upon prior experiential interactions but relies on the presence of institutional cues.	[11,18,52]
Chatbot usage intention	In the context of Chatbot usage intention. The term "intent" pertains to the specific objective or purpose that a customer has in mind while formulating a query or comment.	[43,53]
Customer engagement	The amount of intrinsic drive varies according to surroundings and brand. Different mental, emotional, and behavioral engagement levels with brand interactions characterize this psychological condition.	[14,54]
Loyalty	Loyalty is defined as a deeply held commitment to re-usage of a specific service.	[55] & [44]

Table:2
Cronbach's α , AVE.

Construct Name	Label Name	Cronbach's α	CR	Average variance extracted (AVE)
Compatibility	CM	0.773	0.894	0.741
Perceived ease of use	PU	0.765	0.799	0.615
Performance expectancy	PE	0.816	0.891	0.680
Social influence	SI	0.807	0.869	0.731
Customer engagement	CE	0.846	0.813	0.779
Chatbot initial trust	CT	0.841	0.897	0.686
Chatbot usage intention	CI	0.723	0.798	0.694
Loyalty	L	0.720	0.813	0.653

emphasizes the

predictive capacity of antecedents of chatbot trust (Fig-1), and how it interns Chatbot usage intention, loyalty, and customer engagement. Hence, the partial least squares (PLS) method has been applied in this study, and the testing and measurement of our structural models was carried out with the aid of the Smart PLS 3.2 software (Fig-2). We carried out a confirmatory factor analysis, also known as a CFA, to establish the validity and reliability of the measurement model.

Cronbach's alpha and composite reliability (CR) were employed to assess internal reliability. Each construct's Cronbach's and CR values were higher than the acceptable threshold of 0.7; the constructs are, Compatibility (0.773), perceived Usage (0.765), Performance expectancy (0.816), Social influence (0.807), Customer engagement (0.846), Chatbot initial trust (0.841) chatbot intention (0.723) and loyalty (0.720) All the constructs were loaded as more than threshold value, as per [56] and [51], Discriminant validity was successfully established, as indicated by fulfilling the Fornell-Larcker criterion. Subsequently, the measurement model's convergent validity was confirmed by assessing the factor loading value and the average variance extracted (AVE) values (Table 1) for each item and construct, respectively. All the AVE values more than 0.50, [51].

4.1. Measurement model result

4.2. Results & discussion

4.2.1. Mean, STDEV, T-Values, P-Values

Table 3 indicates that seven out of the proposed six hypotheses were deemed valid, while one of the proposed hypotheses was considered invalid. Chatbot trust has a significantly positive impact on the Perceived ease of use, Performance expectancy, and social influence as indicated by the statistical analysis (H1: $\beta = 0.637$, $t = 16.096$, (H3: $\beta = 0.637$, $t = 16.096$, (H4: $\beta = 0.637$, $t = 16.096$. The results of the study coincide with [8,16,17,41], and [3]. While Compatibility did not have a positive influence on Chatbot trust (H2: $\beta = -0.007$, $t = 0.069$).

4.2.2. Fornell-Larcker Criterion

Table. 4 highlighted the Fornell-Larcker Criterion which indicates the discriminant validity between the latent constructs. Regarding chatbots, trust plays a crucial role in influencing client engagement, loyalty, and usage intention. The conceptual model has also examined how respondents' gender and age moderate the relationship between chatbot trust and customer usage intention. A

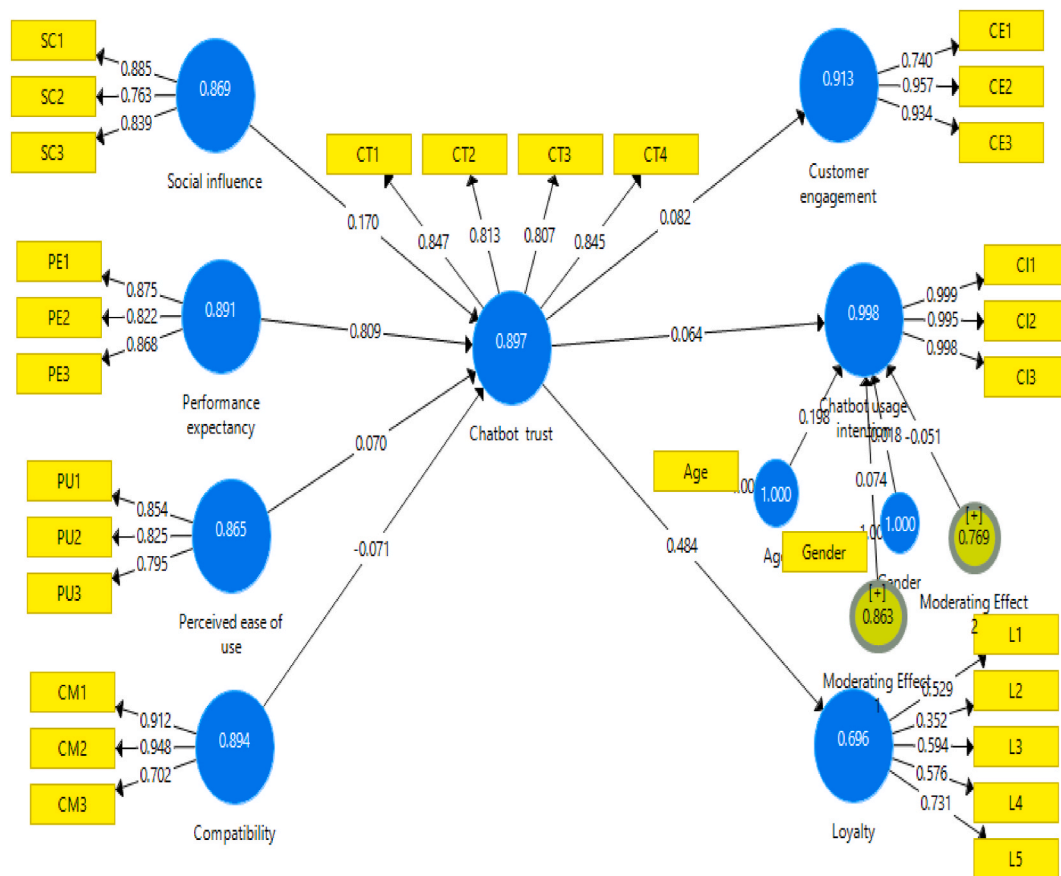


Fig.2. PLS-Structural equation Measurement model result (Pleaser refer to Table 2 above for the full forms of the labels).

Table:3
Mean, STDEV, T-Values, P-values).

Construct	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values	Results Supported YES/NO
$C T_{-} \rightarrow C I_{-}$	0.064	0.072	0.064	0.998	0.001	YES
$C T_{-} \rightarrow C E_{-}$	0.082	0.077	0.103	0.797	0.003	YES
$C T_{-} \rightarrow L$	0.484	0.498	0.051	9.552	0.000	YES
$C_{-} \rightarrow C T_{-}$	0.071	0.073	0.043	1.649	0.231	NO
$P U_{-} \rightarrow C T_{-}$	0.070	0.075	0.049	1.434	0.002	YES
$P E_{-} \rightarrow C T_{-}$	0.809	0.807	0.042	19.078	0.000	YES
$S I_{-} \rightarrow C T_{-}$	0.170	0.169	0.068	2.514	0.012	YES
$G_{-} \rightarrow C I_{-}$	-0.018	-0.016	0.067	0.269	0.004	YES
$A_{-} \rightarrow C I_{-}$	0.198	0.189	0.029	6.844	0.000	YES
Moderating Effect 1-A -> C I_{-}	0.074	0.067	0.112	0.664	0.001	YES
Moderating Effect 2-G -> C I_{-}	-0.051	-0.054	0.038	1.346	0.179	NO

customer's age significantly affects their inclination to use a chatbot, but gender had no discernible moderating effect, according to the study's results. It has evidence in studies by Refs. [8,58].

4.3. Results of R^2

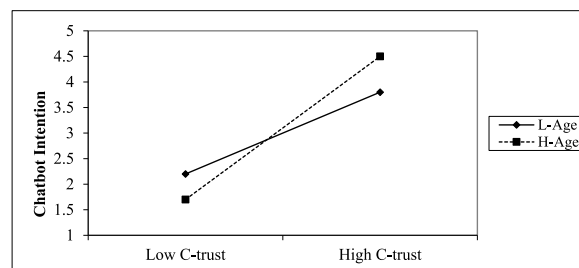
In Table 5, The coefficients of determination (R^2) for the three endogenous constructs (behavioral intention, perceived trust, and use) accounted for a significant proportion of the total variance ($R^2 = 0.4008$ for chatbot trust, $R^2 = 0.902$ for Chatbot usage intention, and $R^2 = 0.239$ loyalty for the use of chatbot). According to [59], the r^2 value is more than 0.2 and is considered substantial. The PLS-

Table:4
Fornell-Larcker criterion.

Fornell-Larcker Criterion								
	CT	CUI	CM	CE	LY	PEU	PE	SI
CT	0.828							
CUI	0.401	0.699						
CM	0.545	0.696	0.757					
CE	0.433	0.277	0.724	0.619				
LY	0.488	0.299	0.246	0.247	0.624			
PEU	0.943	0.232	0.538	0.453	0.463	0.855		
PE	0.608	0.246	0.580	0.423	0.396	0.572	0.825	
SI	0.791	0.277	0.700	0.607	0.424	0.773	0.667	0.831

SEM results in [Table-5](#) demonstrate that the proposed model is statistically significant.

4.4. Moderation effect



The results revealed a significant moderating role of the age group between (16–20), and (21–26) on the relationship between chatbot trust and chatbot intention to use. The plot shows a steeper and positive gradient when the age of the respondents increases. Thus, this indicates that the impact of age in fostering collaborative chatbot intention is more substantial in the age group (21–26) as compared to the intentions of those in the age group (16–20). At the same time, there is a significant moderating impact of gender between chatbot trust and chatbot intention.

The coefficient of determination ([Table 6](#)), sometimes known as the R2 value or the R2 value itself, is a statistical tool that measures the extent to which shifts in one variable may explain shifts in another. The results presented in [Table 5](#) demonstrate that the statistical measure provides valuable insight into the regression model. The value (R2), also known as the coefficient of determination (R2) value, indicates the proportion of the total variance in a dependent variable that can be attributed to the effects of the independent variables. It is referred to as “a measure of the model’s predictive accuracy derived as the squared correlation between key factors on chatbot trust and outcome of chatbot trust such as customer usage intention, loyalty and customer engagement along with moderators. This model has more predictive power ” In other words, it is a correlation that has been squared.

Finally, all the Hypotheses of the study are accepted except [H2](#), i.e., Compatibility has a significantly positive impact on trust to use chatbots.

5. Conclusion

The study emphasizes the significance of ‘trust in chatbots’ and the factors influencing this trust, customer usage intention, loyalty, and customer engagement, and the gender and age of the respondents considered as moderators of the measurement models, according to the study. Consumers’ opinions of brands that offer chatbots could shift depending on how positive or negative their experiences with chatbots are. Based on these results, businesses should develop plans to increase faith in messaging chatbots among customers. Finally, all the hypotheses of the study are accepted except that compatibility has a significantly positive impact on trust in using chatbots.

Table: 5
R square and R square Adjusted.

	R Square	R Square Adjusted
Chatbot trust_	0.902	0.900
Chatbot usage intention_	0.400	0.360
Customer engagement_	0.188	0.184
Loyalty	0.239	0.235

Table:6
Blind fold Values.

	SSO	SSE	Q ² (=1-SSE/SSO)
Chatbot trust	864.000	340.979	0.605
Chatbot usage intention	648.000	639.800	0.013
Compatibility	648.000	648.000	
Customer engagement	648.000	611.025	0.057
Loyalty	864.000	799.902	0.074
Perceived ease of use	648.000	648.000	
Performance expectancy	648.000	648.000	
Social Influence	648.000	648.000	

The current study examining the role of age and gender in having trust in technology found that as age increases, educated young adults are more likely to use mobile messaging chatbots and have a favorable view of the technology. Messenger chatbots for mobile commerce will gain traction with consumers. Brands might view it as the future of e-commerce and an opportunity to expand into new markets. Since most consumers are avid internet users, social media and messenger chatbots present excellent opportunities for commercial expansion. Companies are beginning to use chatbots for customer care, and brands stand to gain from messenger chatbots, which make it easier for customers to find and buy things online. The results of the research can be divided into three fundamental aspects. Initial factors that significantly impact trust in a chatbot are its perceived utility, anticipated level of performance, and social influence. Furthermore, a strong correlation exists between chatbot trust and customer loyalty, intention to use the chatbot and customer expectations. As a result, consumers develop confidence in AI chatbots that enhance the efficiency of the service sector. Second, the need to tailor the service offerings to individual customers is highlighted as a critical factor in establishing credibility. Third, chatbots are a helpful tool for service providers to improve their relationships with customers. The service industry, service providers, and marketing directors benefit from these insights and the findings as they have significant business consequences.

Limitations and future research

Since AI Chatbots are still relatively new in the marketing arena and are continually being developed by academics and practitioners, more research is necessary to comprehend their appropriate application.

Initial trust in chatbots leads to brand engagement. Users could be encouraged to visit the website often, blogging or writing reviews which would give visibility to the brand. Similarly, user interaction on social media could be encouraged.

Some limitations should be taken into consideration while interpreting the research. Data was gathered from Private Higher Education Institutions ranked in the top 3 as per standards of HEI accreditation in the state of Andhra Pradesh, India. The results cannot be broadly applied the world over. Therefore, verifying the proposed links with data from different nations could broaden and improve future studies. Furthermore, as technology is still in its infancy, customers' opinions on such tools tend to change over time. This suggests that the cross-sectional approach limits the study's results. Further studies incorporating a greater range of demographic variety would enhance the richness of this research topic. Nevertheless, the findings of such studies could serve as a foundation for developing and enhancing efficient chatbots in the field of e-commerce. Additionally, they can serve as a valuable resource for managers, e-marketers, and chatbot architects in comprehending the association between trust in chatbots and customers' usage intention, loyalty, and customer engagement. This research model has further scope for expansion in the future. Other variables like customer attitude, and customer satisfaction with the chatbots could be added to understand how initial trust affects the results.

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Data availability statement

Data Available at <https://github.com/venkatchetan/AI-files-Related-Chatbot-repository-Files>.

CRediT authorship contribution statement

Jyothsna M: Writing – original draft, Conceptualization. **Venkata Subbaiah P:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Natalia Kryvinska:** Writing – review & editing, Resources, Methodology, Investigation.

Declaration of competing interest

The main aim of this study is to determine which features of chatbots influence user acceptability, confidence, and engagement. The intention is to develop a model that integrates theories related to initial trust, customer engagement, loyalty, and chatbot adoption. The UTAUT and the DOI paradigm, the TAM Model, and the writings of other writers will serve as the foundation for this paradigm.

Using the PLS-SEM method, we computed the suggested model. Seven predicted links are shown. Out of the six hypotheses that were proposed, seven were determined to be plausible, while one was completely rejected. The study also found that Perceived ease of use, performance expectations, and social influence of chatbots are all greatly enhanced by trust. The statistical analysis also showed that consumers' intentions, loyalty, and engagement levels are highly influenced by their level of trust in chatbots. The association between chatbot trust and intention is moderated by both age and gender. It was shown that respondents' ages had a substantial impact on their plans to use chatbots, but their gender had no effect on those intentions.

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